Statistica

Appelli svolti e archivio dei testi

Parte I: tre assegnazioni del fascicolo del 19 dicembre 2024, con gli svolgimenti personali gia' disponibili.

Appendice: altri appelli storici e materiali per esercitarsi, conservati come testi originali. Le soluzioni non sono presenti nelle fonti di questa appendice.

Parte II: non sono state trovate ulteriori prove generate per questo blocco di inferenza. I venti mock di simulazione appartengono al volume Numerical.

Parte I

Prove passate con svolgimenti

Fascicolo del 19 dicembre 2024. Gli svolgimenti sono appunti personali, non soluzioni ufficiali certificate.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

December 19, 2024

Time allowed: 1 hour and 15 minutes. PLEASE, make an effort to write in a legible and organized way.

1 Assignment 1

Exercise 1

Present the asymptotic theory of (statistical) hypothesis testing for both regular and non-regular parametric models (including at least one example of non-regular models).

Exercise 2

Let $(X_1, \ldots, X_n)$ be a random sample from a Beta distribution with parameter $(\theta, 1)$, $\theta > 0$. Provide a Gaussian limit for the moment-based and the maximum likelihood estimators of θ, and discuss the (large n) asymptotic relative efficiency of these estimators.

Exercise 3

Let X and Y be independent random variables distributed as Uniform on $(0, 1)$. Compute the density function of the random variable $Z = X^2 + Y$ on $(1, 2)$.

Solution 1

Asymptotic theory of hypothesis testing (regular and non-regular parametric models) Hypothesis testing assesses whether data are compatible with a null hypothesis H_0 in a parametric model $\{f(\cdot;\theta) : \theta \in \Theta\}$ against H_1. Given a statistic T_n, one calibrates a critical value c_α so that the test has level α,

$$\sup_{\theta\in\Theta_0} \mathbb{P}_\theta(T_n \geq c_\alpha) \leq \alpha,$$

which requires the null distribution of T_n (since c_α is its quantile). In many classical Gaussian settings (and under regularity), exact finite-sample laws are available (Z, t, χ^2, F); in general, exact null distributions are unavailable or too complex, motivating *asymptotic testing*: derive the limiting distribution as $n \to \infty$ and use asymptotic quantiles. Two standard large-sample constructions are the likelihood-ratio approach (Wilks' theorem in regular models) and estimator-based tests such as Wald; in non-regular models, Wilks may fail and the LR limit under H_0 can be non-χ^2, so we present a regular illustration and a non-regular counterexample.

Wilks' theorem (1D, simple two-sided test). Let $(x_1, \ldots, x_n)$ be a random sample from a *regular* one-dimensional parametric model

$$\{f_\theta : \ \theta \in \Theta \subset \mathbb{R}\},$$

with Θ open. Consider testing the simple two-sided hypothesis

$$H_0 : \ \theta = \theta_0 \qquad \text{vs} \qquad H_1 : \ \theta \neq \theta_0,$$

where $\theta_0 \in \Theta$. Define the likelihood ratio statistic

$$\Lambda(x_1, \ldots, x_n) = \frac{L(\theta_0 \,|\, x)}{\sup_{\theta\in\Theta} L(\theta \,|\, x)}.$$

Then under H_0 and as $n \to \infty$,

$$-2\log\Lambda(x_1, \ldots, x_n) \ \xrightarrow{d} \ \chi^2_1.$$

Hence an asymptotic level-α LRT rejects when

$$-2\log\Lambda(x_1, \ldots, x_n) \ \geq \ \chi^2_{1,\,1-\alpha}.$$

Proof. Let $\ell_n(\theta) := \log L(\theta; x)$ and let $\hat\theta_n$ be the MLE.

$$\log\Lambda = \log L(\theta_0; n) - \log L(\hat\theta_n; n) = \ell_n(\theta_0) - \ell_n(\hat\theta_n).$$

$$-2\log\Lambda = -2\ell_n(\theta_0) + 2\ell_n(\hat\theta_n).$$

We now use a second-order Taylor approximation of $\ell_n(\theta_0)$ around $\hat\theta_n$.

$$-2\log\Lambda = -2\Big[\underbrace{\ell_n(\hat\theta_n)}_{A} + \underbrace{\ell_n'(\hat\theta_n)(\theta_0 - \hat\theta_n)}_{B} + \underbrace{\frac{1}{2}\,\ell_n''(\hat\theta_n)(\theta_0 - \hat\theta_n)^2}_{C} + \underbrace{r_n}_{D}\Big] + \underbrace{2\ell_n(\hat\theta_n)}_{E}.$$

$$A + E = 0, \quad B = 0 \quad \text{since } \hat\theta_n \text{ maximizes } \ell_n(\theta) \text{ so } \ell_n'(\hat\theta_n) = 0.$$

Since we are interested in studying the asymptotic behaviour when $n \to \infty$, the r_n (D) is negligible. Hence

$$-2\log\Lambda \;\simeq\; -\ell_n''(\hat\theta_n)\,(\theta_0-\hat\theta_n)^2 \qquad \text{when } n\to+\infty.$$

$$-2\log\Lambda \simeq \underbrace{-\frac{1}{n}\,\ell_n''(\hat\theta_n)}_{a}\cdot\underbrace{n(\theta_0-\hat\theta_n)^2}_{b}.$$

$$a=-\frac{1}{n}\,\ell_n''(\hat\theta_n) \;\xrightarrow[n\to\infty]{\mathbb{P}}\; I(\theta_0), \qquad \sqrt{n}\,(\hat\theta_n-\theta_0) \;\xrightarrow[n\to\infty]{d}\; \mathcal{N}\!\left(0,\frac{1}{I(\theta_0)}\right).$$

Therefore

$$b=n(\theta_0-\hat\theta_n)^2=\big(\sqrt{n}(\hat\theta_n-\theta_0)\big)^2 \;\xrightarrow[n\to\infty]{d}\; \frac{1}{I(\theta_0)}\,Z^2, \qquad Z\sim\mathcal{N}(0,1).$$

By Slutsky's theorem,

$$-2\log\Lambda \;\xrightarrow[n\to\infty]{d}\; I(\theta_0)\cdot\frac{1}{I(\theta_0)}\,Z^2=Z^2\sim\chi^2_1,$$

so the limit distribution is χ^2_1 (and in general constrained case χ^2_r). □

Poisson LRT example (regular case). Let $(X_1,\dots,X_n)$ be a random sample from $X\sim$ Poisson(λ), $\lambda>0$, and test

$$H_0:\lambda=\lambda_0 \qquad \text{vs} \qquad H_1:\lambda\neq\lambda_0.$$

The likelihood is

$$L(\lambda;x)=\prod_{i=1}^n\frac{e^{-\lambda}\lambda^{x_i}}{x_i!}=\frac{e^{-n\lambda}\lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}.$$

The unrestricted MLE is $\widehat\lambda_n=\bar X_n$, and the LR statistic is

$$\Lambda=\frac{L(\lambda_0;x)}{L(\widehat\lambda_n;x)}=\exp\big(-n(\lambda_0-\widehat\lambda_n)\big)\left(\frac{\lambda_0}{\widehat\lambda_n}\right)^{\sum_{i=1}^n x_i}.$$

Using $\sum_{i=1}^n x_i=n\widehat\lambda_n$, we obtain

$$-2\log\Lambda=2n\left[\widehat\lambda_n\log\left(\frac{\widehat\lambda_n}{\lambda_0}\right)-(\widehat\lambda_n-\lambda_0)\right].$$

Under H_0, Wilks' theorem yields

$$-2\log\Lambda\xrightarrow{d}\chi^2_1 \qquad (n\to\infty),$$

so an asymptotic level-α test rejects when $-2\log\Lambda\geq\chi^2_{1,\,1-\alpha}$.

Non-regular models: Wilks may fail (example). Let $X_1,\dots,X_n$ be i.i.d. with density

$$f(x;\theta)=e^{-(x-\theta)}\mathbf{1}\{x\geq\theta\}, \qquad \theta\in\mathbb{R}.$$

Testing $H_0:\theta=\theta_0$ vs $H_1:\theta\neq\theta_0$, the likelihood is proportional to $e^{-\sum_{i=1}^n(x_i-\theta)}\mathbf{1}\{\theta\leq X_{(1)}\}$, hence the MLE is $\widehat\theta_n=X_{(1)}$. One finds

$$-2\log\Lambda=2n(\widehat\theta_n-\theta_0)=2n(X_{(1)}-\theta_0).$$

Under H_0, $n(X_{(1)} - \theta_0) \sim \text{Exp}(1)$, so

$$-2\log\Lambda \sim \text{Exp}\left(\tfrac{1}{2}\right),$$

which is not χ^2. This failure is due to non-regularity: the support depends on θ (a boundary issue), violating the regularity conditions behind Wilks; in regular models, however, one can also build asymptotic tests directly from the Gaussian limit of an estimator, leading to the Wald approach.

Wald test (asymptotic, regular models). Assume an estimator T_n of θ satisfies asymptotic normality

$$\sqrt{n}\,(T_n - \theta) \xrightarrow{d} \mathcal{N}(0, V), \qquad V > 0,$$

and we have a consistent standard error S_n such that $S_n/\sigma_n \xrightarrow{p} 1$, where $\sigma_n = \sqrt{\text{Var}(T_n)}$. Then for testing $H_0 : \theta = \theta_0$ one uses

$$z_n = \frac{T_n - \theta_0}{S_n} \xrightarrow{d} \mathcal{N}(0,1), \qquad n \to \infty,$$

and rejects for large deviations from 0: in the two-sided case $H_1 : \theta \neq \theta_0$, reject when $|z_n| \geq z_{1-\alpha/2}$ (analogously, one-sided tests reject when $z_n \geq z_{1-\alpha}$ or $z_n \leq z_\alpha$).

Example (Bernoulli, Wald). For $X_i \sim \text{Bernoulli}(\theta)$, $T_n = \hat{\theta}_n = \frac{1}{n}\sum_{i=1}^n X_i$ and

$$S_n^2 = \frac{1}{n}\,\hat{\theta}_n(1 - \hat{\theta}_n).$$

Then

$$z = \frac{\hat{\theta}_n - \theta_0}{\sqrt{\frac{1}{n}\,\hat{\theta}_n(1 - \hat{\theta}_n)}} \xrightarrow{d} \mathcal{N}(0,1),$$

and for $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$ one rejects when $|z| > z_{1-\alpha/2}$.

Solution 2

Setup.

$$\text{Let } X_1, \ldots, X_n \sim \text{Be}(\theta, 1).$$

Recall (general Beta moments):

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}, \qquad \text{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

Here $\alpha = \theta,\ \beta = 1$, hence

$$\mathbb{E}[X] = \frac{\theta}{\theta + 1}, \qquad \text{Var}(X) = \frac{\theta}{(\theta+1)^2(\theta+2)}.$$

Density and CDF (as written).

$$f_X(x;\theta,1) = \frac{\Gamma(\theta+1)}{\Gamma(\theta)\Gamma(1)} x^{\theta-1}(1-x)^0 = \theta x^{\theta-1}\mathbf{1}_{(0,1)}(x).$$

Also

$$F_X(x) = \int_0^x \theta t^{\theta-1}\, dt = \begin{cases} 0, & x < 0, \\ x^\theta, & x \in [0,1], \\ 1, & x > 1. \end{cases}$$

MLE (your scheme).

$$L_X(x;\theta) = \prod_{i=1}^n \theta x_i^{\theta-1} = \theta^n \prod_{i=1}^n x_i^{\theta-1},$$

$$\ell_X(\theta) = \log L_X(\theta) = n\ln\theta + (\theta-1)\sum_{i=1}^n \ln(x_i).$$

Differentiate and set to zero:

$$\frac{\partial}{\partial\theta}\ell_X(\theta) = \frac{n}{\theta} + \sum_{i=1}^n \ln(x_i) = 0 \quad \Longrightarrow \quad \hat\theta_{\text{MLE}} = -\frac{n}{\sum_{i=1}^n \ln(x_i)}.$$

Transformation $Y = \ln(X)$, then $Z = -Y$ (your steps).

Let $Y = \ln(X)$. For $y \in \mathbb{R}$,

$$\mathbb{P}(Y \le y) = \mathbb{P}(\ln X \le y) = \mathbb{P}(X \le e^y) = \begin{cases} 0, & e^y < 0, \\ e^{\theta y}, & e^y \in [0,1] \ \Leftrightarrow\ y \le 0, \\ 1, & e^y > 1 \ \Leftrightarrow\ y > 0, \end{cases}$$

so

$$f_Y(y) = \theta e^{\theta y}\mathbf{1}_{\{y\le 0\}}.$$

Now set $Z = -Y$. Then for $z \ge 0$,

$$\mathbb{P}(Z \le z) = \mathbb{P}(Y \ge -z) = 1 - e^{-\theta z},$$

hence

$$f_Z(z) = \theta e^{-\theta z}\mathbf{1}_{\{z\ge 0\}}, \qquad Z \sim \text{Exp}(\theta),$$

$$\mathbb{E}[Z] = \frac{1}{\theta}, \qquad \text{Var}(Z) = \frac{1}{\theta^2}.$$

For the transformed sample $Z_i = -\ln(X_i)$,

$$L_Z(z;\theta) = \theta^n \exp\Big(-\theta\sum_{i=1}^n z_i\Big)\,\mathbf{1}_{\{\min_i z_i \ge 0\}},$$

$$\ell_Z(\theta) = n\ln\theta - \theta\sum_{i=1}^n z_i.$$

Score:

$$\frac{\partial}{\partial\theta}\ell_Z(\theta) = \frac{n}{\theta} - \sum_{i=1}^n z_i = 0 \quad \Longrightarrow \quad \hat\theta_{\text{MLE}} = \frac{n}{\sum_{i=1}^n z_i} = \frac{1}{\bar Z_n}.$$

Gaussian limit for the MLE (your CLT + Delta method route).

By CLT:

$$\sqrt{n}(\bar{Z}_n - \mathbb{E}[Z]) \Rightarrow \mathcal{N}(0, \mathrm{Var}(Z)) \quad \Longrightarrow \quad \sqrt{n}\Big(\bar{Z}_n - \frac{1}{\theta}\Big) \Rightarrow \mathcal{N}\Big(0, \frac{1}{\theta^2}\Big).$$

Let $g(t) = 1/t$. Then $g'(t) = -1/t^2$. With $\mu = \mathbb{E}[Z] = 1/\theta$,

$$g'(\mu) = -\frac{1}{(1/\theta)^2} = -\theta^2, \qquad (g'(\mu))^2 = \theta^4.$$

Delta method:

$$\sqrt{n}\big(g(\bar{Z}_n) - g(\mu)\big) \Rightarrow \mathcal{N}\Big(0,\ (g'(\mu))^2 \mathrm{Var}(Z)\Big) = \mathcal{N}\Big(0, \theta^4 \cdot \frac{1}{\theta^2}\Big) = \mathcal{N}(0, \theta^2).$$

So

$$\sqrt{n}(\hat{\theta}_{\mathrm{MLE}} - \theta) \Rightarrow \mathcal{N}(0, \theta^2).$$

Method of moments (your scheme).

$$m_1 = \bar{X}_n, \qquad \mu_1 = \mathbb{E}[X] = \frac{\theta}{\theta+1}, \qquad m_1 = \mu_1.$$

Solve:

$$\bar{X}_n = \frac{\theta}{\theta+1} \iff \bar{X}_n = 1 - \frac{1}{\theta+1} \iff \frac{1}{\theta+1} = 1 - \bar{X}_n \iff \theta + 1 = \frac{1}{1-\bar{X}_n}$$

$$\Longrightarrow \quad \hat{\theta}_{\mathrm{MOM}} = \frac{1}{1-\bar{X}_n} - 1 = \frac{\bar{X}_n}{1-\bar{X}_n}.$$

Let $g(m) = \frac{m}{1-m}$. Then

$$g'(m) = \frac{(1-m) - m(-1)}{(1-m)^2} = \frac{1}{(1-m)^2}.$$

Also

$$g'(\mu) = \left(\frac{1}{1 - \frac{\theta}{1+\theta}}\right)^2 = \left(\frac{1}{\frac{1}{1+\theta}}\right)^2 = (1+\theta)^2.$$

CLT:

$$\sqrt{n}(\bar{X}_n - \mathbb{E}[X]) \Rightarrow \mathcal{N}(0, \mathrm{Var}(X)).$$

Delta method:

$$\sqrt{n}(\hat{\theta}_{\mathrm{MOM}} - \theta) \Rightarrow \mathcal{N}\Big(0,\ (g'(\mu))^2 \mathrm{Var}(X)\Big) = \mathcal{N}\left(0,\ (1+\theta)^4 \cdot \frac{\theta}{(1+\theta)^2(2+\theta)}\right) = \mathcal{N}\left(0,\ \frac{(1+\theta)^2\theta}{2+\theta}\right).$$

ARE (your algebra).

$$\mathrm{ARE}(\hat{\theta}_{\mathrm{MOM}}, \hat{\theta}_{\mathrm{MLE}}) = \frac{\frac{(1+\theta)^2\theta}{2+\theta}}{\theta^2} = \frac{(1+\theta)^2}{(2+\theta)\theta} = \frac{\theta^2 + 2\theta + 1}{\theta^2 + 2\theta} = 1 + \frac{1}{\theta^2 + 2\theta} > 1.$$

Hence $\hat{\theta}_{\mathrm{MLE}}$ is more efficient if $\theta > 0$.

Solution 3

$$z = x^2 + y \qquad X \sim U(0,1) \qquad Y \sim U(0,1)$$

$$z \in [0,2] \qquad U = X^2$$

$$\mathbb{P}(Z = z) = \mathbb{P}\big(X^2 = x,\ Y = z - x\big) = \int f_U(x)\, f_Y(z-x)\, dx \Rightarrow$$

$$\mathbb{P}(U \le x) = \mathbb{P}(X^2 \le x) = \mathbb{P}(-\sqrt{x} \le X \le \sqrt{x}) = \mathbb{P}(X \le \sqrt{x}) = \begin{cases} 0 & x < 0 \\ \sqrt{x} & x \in (0,1] \\ 1 & x > 1 \end{cases}$$

$$F_X(x) = \begin{cases} 0 & x < 0 \\ x & x \in (0,1) \\ 1 & x > 1 \end{cases} \qquad f_U(x) = \frac{1}{2\sqrt{x}}\, \mathbf{1}_{\{x \in [0,1]\}}$$

$$0 \le y \le 1 \qquad 0 \le x \le 1 \qquad -z \le x - z \le 1 - z$$

$$x \in [0,1] = A \qquad z - x \in [z-1, z] = B$$

$$= \int_{\max\{z-1,0\}}^{\min\{z,1\}} \frac{1}{2}\, \frac{1}{\sqrt{x}}\, dx = \int_{\max(z-1,0)}^{\min(z,1)} \frac{1}{2}\, x^{-1/2}\, dx = x^{1/2}\Big|_{\max(z-1,0)}^{\min(z,1)}$$

$$h_z(z) = \begin{cases} 0 & z < 0 \\ \sqrt{z} & z \in [0,1] \\ 1 - \sqrt{z-1} & z \in (1,2] \\ 0 & z > 2 \end{cases}$$

2 Assignment 2

Exercise 1

Present the large-sample asymptotic properties of maximum likelihood estimators within the class of regular statistical models, with focus on Cramér theorem and its proof.

Exercise 2

Let (X_1, X_2) be a random sample from a Beta distribution with parameter $(\theta, 1)$, with unknown $\theta > 0$. To test $H_0 : \ \theta \leq 1$ versus $H_1 : \ \theta > 1$, consider the rejection region

$$C = \{(x_1, x_2) \in (0,1)^2 : \ x_2 \geq \sqrt{x_1}\}$$

and find the power function

$$\beta(\theta) = \mathbb{P}_\theta\big((X_1, X_2) \in C\big).$$

Find the most powerful test of size $\alpha = 3/8$ for testing $H_0 : \ \theta = 1$ versus $H_1 : \ \theta = 3$.

Exercise 3

A disease affects 1% of a population. A diagnostic test correctly identifies the disease in 95% of infected individuals and incorrectly gives a positive result in 5% of healthy individuals. If a person tests positive, what is the probability that they actually have the disease?

Solution 1

3) Asympt. efficiency = w.c. + Cramer–Rao (Consequences: invariance, etc.)

Let $X_1, \dots, X_n \sim F_\theta, \quad \theta \in \Theta$ **REGULAR** model

Let $\hat{\theta}_n$ be the maximum likelihood estimator for θ.

If $\hat{\theta}_n$ is weakly consistent $\hat{\theta}_n \xrightarrow{p} \theta_0$

$$\Big(\text{i.e. } \lim_{n\to\infty} \mathbb{P}\big(|\hat{\theta}_n - \theta_0| > \varepsilon\big) = 0\Big)$$

then

$$\sqrt{n}\,(\hat{\theta}_n - \theta_0) \xrightarrow{d} \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right) \qquad \text{for } n \to \infty.$$

Dim

If the model is regular then

$$\left|\frac{\partial}{\partial\theta} f(x;\theta)\right| \le g(x), \qquad \left|\frac{\partial^2}{\partial\theta^2} f(x;\theta)\right| \le h(x), \qquad \left|\frac{\partial^3}{\partial\theta^3} \log f(x;\theta)\right| \le H(x)$$

for suitable positive, integrable functions $g(\cdot)$, $h(\cdot)$, and $H(\cdot)$ (i.e. $\mathbb{E}[H(X)] < +\infty$).

Let $\ell(\theta) = \log f(X_1, \dots, X_n; \theta)$ be the log likelihood function.

If we expand $\ell'(\theta)$ w.r.t. Taylor and Lagrange remainder:

$$\ell'(\theta) = \ell'(\theta_0) + \ell''(\theta_0)(\theta - \theta_0) + \frac{1}{2}\ell^{(3)}(\bar{\theta})(\theta - \theta_0)^2, \qquad \bar{\theta} \in [\theta_0, \theta].$$

By replacing θ with $\hat{\theta}$:

$$\begin{aligned} 0 = \ell'(\hat{\theta}) &= \ell'(\theta_0) + \ell''(\theta_0)(\hat{\theta} - \theta_0) + \frac{1}{2}\ell^{(3)}(\tilde{\theta})(\hat{\theta} - \theta_0)^2, \qquad \tilde{\theta} \in [\hat{\theta}, \theta_0] \\ &= \ell'(\theta_0) + (\hat{\theta} - \theta_0)\Big[\ell''(\theta_0) + \frac{1}{2}\ell^{(3)}(\tilde{\theta})(\hat{\theta} - \theta_0)\Big]. \end{aligned}$$

Hence

$$\sqrt{n}(\hat{\theta} - \theta_0) = -\frac{\frac{1}{\sqrt{n}}\ell'(\theta_0)}{\frac{1}{n}\ell''(\theta_0) + \frac{1}{2n}\ell^{(3)}(\tilde{\theta})(\hat{\theta} - \theta_0)}.$$

Let's study $-\dfrac{1}{\sqrt{n}}\ell'(\theta_0)$:

$$-\frac{1}{\sqrt{n}}\ell'(\theta_0) = -\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{\partial}{\partial\theta}\log f(x_i;\theta_0) = -\frac{1}{\sqrt{n}}\sum_{i=1}^{n}\frac{\frac{\partial}{\partial\theta}f(x_i;\theta_0)}{f(x_i;\theta_0)}.$$

By defining

$$W := -\sum_{i=1}^{n}\frac{\frac{\partial}{\partial\theta}f(x_i;\theta_0)}{f(x_i;\theta_0)},$$

we can say that $\mathbb{E}(W) = 0 \quad \mathrm{Var}(W) = I_n(\theta_0)$

$$\frac{1}{\sqrt{n}} W \overset{w}{\sim} \mathcal{N}(0, I(\theta_0)) .$$

Let's study $\frac{1}{n}\ell''(\theta_0)$:

$$\begin{aligned}\frac{1}{n}\ell''(\theta_0) &= \frac{1}{n}\sum_{i=1}^{n} \frac{\partial^2}{\partial\theta^2} \log f(x_i;\theta_0) \\ &= \frac{1}{n}\sum_{i=1}^{n} \frac{\frac{\partial^2}{\partial\theta^2} f(x_i;\theta_0)}{f(x_i;\theta_0)} - \frac{1}{n}\sum_{i=1}^{n} \left(\frac{\frac{\partial}{\partial\theta} f(x_i;\theta_0)}{f(x_i;\theta_0)} \right)^2 .\end{aligned}$$

We know that

$$\frac{1}{n}\sum_{i=1}^{n} \frac{\frac{\partial^2}{\partial\theta^2} f(x_i;\theta_0)}{f(x_i;\theta_0)} \to 0 \qquad \text{for LLN}$$

and

$$\frac{1}{n}\sum_{i=1}^{n} \left(\frac{\frac{\partial}{\partial\theta} f(x_i;\theta_0)}{f(x_i;\theta_0)} \right)^2 \to I(\theta_0).$$

Therefore

$$\frac{1}{n}\ell''(\theta_0) \to -I(\theta_0).$$

Let's study $\frac{1}{2n}\ell^{(3)}(\tilde{\theta})(\hat{\theta} - \theta_0)$.

We know that

$$\frac{1}{n}\ell^{(3)}(\tilde{\theta}) = \frac{1}{n}\sum_{i=1}^{n} \left| \frac{\partial^3}{\partial\theta^3} \log f(x_i;\tilde{\theta}) \right| \leq \frac{1}{n}\sum_{i=1}^{n} H(x_i) \in \mathbb{E}[H(X_1)] < +\infty \quad \text{finite.}$$

Therefore $\frac{1}{n}\ell^{(3)}(\tilde{\theta}) = o(1)$.
Moreover $\hat{\theta} - \theta_0 \ = o(1)$.
Therefore $\frac{1}{n}\ell^{(3)}(\tilde{\theta})(\hat{\theta} - \theta_0) \to 0$.
In conclusion

$$\sqrt{n}(\hat{\theta} - \theta_0) \overset{w}{\sim} \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right).$$

Consequences

- MLE is asympt Gaussian $\qquad \sqrt{n}(\hat{\theta} - \theta_0) \to \mathcal{N}\left(0, \frac{1}{I(\theta_0)}\right)$
- MLE is asympt correct $\qquad \mathbb{E}(\hat{\theta}) \to \theta_0$
- MLE is asympt efficient $\qquad \mathrm{Var}(\hat{\theta}) = \frac{1}{n\, I(\theta_0)}$

If we are interested in estimating $g(\theta)$ then by the invariance prop of MLE the MLE of $g(\theta)$ is $g(\hat{\theta})$.

Moreover

$$\sqrt{n}\big(g(\hat{\theta}) - g(\theta)\big) \xrightarrow{d} \mathcal{N}\left(0, \frac{\big(g'(\theta_0)\big)^2}{I(\theta_0)}\right) \qquad \text{for } n \to \infty.$$

Solution 2

$$\text{Let } X_1, X_2 \sim \text{Beta}(\theta, 1), \qquad \theta > 0. \qquad H_0 : \ \theta \leq 1, \qquad H_1 : \ \theta > 1.$$

$$\begin{aligned} f_X(x) &= \frac{\Gamma(\theta+b)}{\Gamma(\theta)\Gamma(b)}\, x^{\theta-1}(1-x)^{b-1}\, \mathbf{1}_{(0,1)}(x) \\ &= \frac{\Gamma(\theta+1)}{\Gamma(\theta)\Gamma(1)}\, x^{\theta-1} = \frac{\Gamma(\theta+1)}{\Gamma(\theta)}\, x^{\theta-1} = \theta x^{\theta-1}\, \mathbf{1}_{[0,1]}(x). \end{aligned}$$

$$L_{\mathbf{x}}(\theta) = \theta^n \prod_{i=1}^{n} x_i^{\theta-1} = \theta^2\, x_1^{\theta-1} x_2^{\theta-1}.$$

$$F_X(x) = \int_0^x \theta t^{\theta-1}\, dt = \begin{cases} 0, & x < 0, \\ x^\theta, & x \in [0,1], \\ 1, & x > 1, \end{cases} \qquad \overline{F}_X(x) = \begin{cases} 1, & x < 0, \\ 1 - x^\theta, & x \in [0,1], \\ 0, & x > 1. \end{cases}$$

$$\begin{aligned} \beta(\theta) &= \mathbb{P}\Big(X_2 \geq \sqrt{X_1}\Big) = \int_0^1 \mathbb{P}_\theta(X_2 \geq \sqrt{x_1})\ f_{X_1}(x_1)\, dx_1 \\ &= \int_0^1 \left(1 - x_1^{\theta/2}\right) \theta x_1^{\theta-1}\, dx_1 = \int_0^1 \theta x^{\theta-1}\, dx - \int_0^1 \theta x^{\frac{3}{2}\theta-1}\, dx \\ &= \int_0^1 \theta x^{\theta-1}\, dx - \frac{2}{3}\int_0^1 \frac{3}{2}\theta x^{\frac{3}{2}\theta-1}\, dx = 1 - \frac{2}{3} = \frac{1}{3} \qquad \forall\, \theta > 0. \end{aligned}$$

$$\int \alpha x^{\alpha-1}\, dx = x^\alpha + c.$$

$$\alpha = \frac{3}{8}, \qquad H_0 : \ \theta = 1, \qquad H_1 : \ \theta = 3.$$

$$\Lambda(\mathbf{x}) = \frac{L_{\mathbf{x}}(\mathbf{x}; \theta_1)}{L_{\mathbf{x}}(\mathbf{x}; \theta_0)} = 3^n \prod_{i=1}^{n} x_i^2 = 3^2 x_1^2 x_2^2 = 9(x_1 x_2)^2.$$

as a monotone non decreasing function of $X_1 X_2 = T(\mathbf{x})$. Under H_0 we reject the null hyp if

$$\mathbb{P}(T(X) > k)\,\Big|_{\theta=\theta_0} = \alpha \quad \Rightarrow \quad \mathbb{P}(X_1 X_2 > k)\,\Big|_{\theta=\theta_0} = \alpha.$$

or equivalently we reject the null hyp if

$$\mathbb{P}\Big(9\,(T(X))^2 > \tilde{k}\Big)\,\Big|_{\theta=\theta_0} = \alpha, \qquad \tilde{k} = \sqrt{k/3}, \qquad \text{because } |XY| = XY.$$

Let $Z = X_1 X_2$. under H_0 $f_{X_1}(x_1) = 1 \cdot x_1^{1-1}\, \mathbf{1}_{[0,1]}(x_1) = \mathbf{1}_{[0,1]}(x_1)$ hence $X_1, X_2 \sim U(0,1)$.

$$U = X_1, \qquad V = X_1 X_2, \qquad x_1 = u, \ \ x_2 = \frac{v}{u}.$$

$$\{0 \le u \le 1\} \cap \Big\{0 \le \frac{v}{u} \le 1\Big\} \;\Rightarrow\; 0 \le v \le u \le 1.$$

$$\begin{aligned} f_{U,V}(u,v) &= f_X\big(g_1^{-1}(u,v), g_2^{-1}(u,v)\big)\,\big|\det J(u,v)\big| \\ &= f_{X_1}\big(g_1^{-1}(u,v)\big)\, f_{X_2}\big(g_2^{-1}(u,v)\big)\cdot\frac{1}{u} \\ &= \mathbf{1}_{(0,1)}(u)\,\mathbf{1}_{(0,1)}\Big(\frac{v}{u}\Big)\cdot\frac{1}{u} = \mathbf{1}_{(v,1)}(u)\,\mathbf{1}_{(0,u)}(v)\cdot\frac{1}{u}. \end{aligned}$$

$$J = \begin{vmatrix} \frac{\partial x_1}{\partial u} & \frac{\partial x_1}{\partial v} \\ \frac{\partial x_2}{\partial u} & \frac{\partial x_2}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -\frac{v}{u^2} & \frac{1}{u} \end{vmatrix} = \frac{1}{u}.$$

$$f_V(v) = \int_v^1 f_{U,V}(u,v)\,du = \int_v^1 \frac{1}{u}\,du = \ln(1) - \ln(v) = -\ln(v).$$

$$\begin{aligned} F_V(v) &= \int_0^v -\ln(t)\,dt = \int_v^0 \ln(t)\,dt = t(\ln t - 1)\Big|_v^0 \\ &= -v(\ln v - 1) = v(1 - \ln v) = v - v\ln v. \end{aligned}$$

$$\int \ln(x)\,dx = x\ln x - \int x\frac{1}{x}\,dx = x(\ln x - 1) + c = x\ln x - x + c, \qquad f = \ln(x),\ f' = \frac{1}{x},\ g' = 1,\ g = x.$$

$$\begin{aligned} \mathbb{P}(T(X) > k) = \alpha &\iff \mathbb{P}(V > k) = \alpha \iff 1 - k + k\ln k = \alpha \\ &\iff k\ln k - k + \frac{5}{8} = 0. \end{aligned}$$

$$\text{We reject } H_0 \iff x_1, x_2 :\ \mathbb{P}(X_1X_2 > k) = \frac{3}{8} \iff k\ln k - k + \frac{5}{8} = 0.$$

Solution 3

$$\mathbb{P}(D) = 0.01 \qquad \mathbb{P}(\overline{D}) = 0.99$$
$$\mathbb{P}(+ \mid D) = 0.95 \qquad \mathbb{P}(+ \mid \overline{D}) = 0.05$$

$$\mathbb{P}(D \mid +)\,?$$

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \;\Rightarrow\; \mathbb{P}(A \cap B) = \mathbb{P}(A \mid B)\,\mathbb{P}(B)$$

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \mid B)\,\mathbb{P}(B)}{\mathbb{P}(A)}$$

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\,\mathbb{P}(D)}{\mathbb{P}(+)} = \frac{0.95 \cdot 0.01}{0.059} \simeq 0.16$$

$$\mathbb{P}(+) = \mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid \overline{D})\mathbb{P}(\overline{D}) = 0.95 \cdot 0.01 + 0.05 \cdot 0.99 = 0.059$$

3 Assignment 3

Exercise 1

Present the asymptotic theory of (statistical) hypothesis testing for both regular and non-regular parametric models (including at least one example of non-regular models).

Exercise 2

Let $(X_1, \ldots, X_n)$ be a random sample from a Gamma distribution with parameters (α, θ), with $\alpha > 0$ known and $\theta > 0$ unknown. Provide a Gaussian limit for the moment-based and the maximum likelihood estimators of θ, and discuss the (large n) asymptotic relative efficiency of these estimators.

Exercise 3

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Compute $\mathbb{E}[X]$.

Solution 1

Asymptotic theory of hypothesis testing (regular and non-regular parametric models) Assignment 1 VEDI SOPRA

Solution 2

Setup (your parametrization).

$$\text{Let } X_1, \dots, X_n \sim \text{Ga}\left(\alpha, \frac{1}{\theta}\right), \qquad \mathbb{E}[X] = \alpha\theta, \qquad \text{Var}(X) = \alpha\theta^2.$$

Density:

$$f_\theta(x) = \frac{(1/\theta)^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-x/\theta}\, \mathbf{1}_{\{x\geq 0\}}.$$

Likelihood.

$$L_X(x;\theta) = \prod_{i=1}^n \frac{(1/\theta)^\alpha}{\Gamma(\alpha)} x_i^{\alpha-1} e^{-x_i/\theta}\, \mathbf{1}_{\{x_i\geq 0\}} = \frac{(1/\theta)^{n\alpha}}{\Gamma(\alpha)^n} \left(\prod_{i=1}^n x_i^{\alpha-1}\right) e^{-(1/\theta)\sum_{i=1}^n x_i}\, \mathbf{1}_{\{\min_i x_i\geq 0\}}.$$

Log-likelihood:

$$\ell_X(\theta) = -n\alpha \ln(\theta) - n \ln \Gamma(\alpha) + (\alpha-1)\sum_{i=1}^n \ln(x_i) - \frac{1}{\theta}\sum_{i=1}^n x_i.$$

MLE (your derivatives).

$$\frac{\partial}{\partial\theta}\ell_X(\theta) = -\frac{n\alpha}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^n x_i.$$

Set to zero:

$$-\frac{n\alpha}{\theta} + \frac{1}{\theta^2}\sum_{i=1}^n x_i = 0 \quad\Longrightarrow\quad \sum_{i=1}^n x_i = n\alpha\theta \quad\Longrightarrow\quad \hat\theta_{\text{MLE}} = \frac{\sum_{i=1}^n x_i}{n\alpha} = \frac{\bar X_n}{\alpha}.$$

Second derivative:

$$\frac{\partial^2}{\partial\theta^2}\ell_X(\theta) = \frac{n\alpha}{\theta^2} - 2\frac{\sum_{i=1}^n x_i}{\theta^3} = -\frac{1}{\theta^2}\left(\frac{2\sum_{i=1}^n x_i}{\theta} - n\alpha\right) = -\frac{n}{\theta^2}\left(\frac{2\bar X_n}{\theta} - \alpha\right).$$

At $\theta = \hat\theta_{\text{MLE}} = \bar X_n/\alpha$,

$$\left.\frac{\partial^2}{\partial\theta^2}\ell_X(\theta)\right|_{\hat\theta_{\text{MLE}}} < 0 \quad\Longleftrightarrow\quad \frac{2\bar X_n}{\hat\theta_{\text{MLE}}} - \alpha > 0,$$

and since

$$\frac{2\bar X_n}{\hat\theta_{\text{MLE}}} - \alpha = \frac{2\bar X_n}{\bar X_n/\alpha} - \alpha = 2\alpha - \alpha = \alpha > 0,$$

it is indeed negative (with $\alpha > 0$).

Gaussian limit (your CLT line).

$$\sqrt{n}(\bar{X}_n - \mathbb{E}[X]) \Rightarrow \mathcal{N}(0, \mathrm{Var}(X)).$$

With $\mathbb{E}[X] = \alpha\theta$ and $\mathrm{Var}(X) = \alpha\theta^2$,

$$\sqrt{n}(\bar{X}_n - \alpha\theta) \Rightarrow \mathcal{N}(0, \alpha\theta^2).$$

Since $\hat{\theta}_{\mathrm{MLE}} = \bar{X}_n/\alpha$,

$$\sqrt{n}\left(\frac{\bar{X}_n}{\alpha} - \theta\right) \Rightarrow \mathcal{N}\left(0, \frac{1}{\alpha^2}\alpha\theta^2\right) = \mathcal{N}\left(0, \frac{\theta^2}{\alpha}\right).$$

Equivalently,

$$\sqrt{n}(\hat{\theta}_{\mathrm{MLE}} - \theta) \Rightarrow \mathcal{N}\left(0, \frac{\theta^2}{\alpha}\right).$$

Metodo dei momenti (your scheme).

$$m_1 = \frac{1}{n}\sum_{i=1}^{n} X_i = \bar{X}_n, \qquad \mu_1 = \mathbb{E}[X] = \alpha\theta, \qquad \bar{X}_n = \alpha\theta.$$

Hence

$$\hat{\theta}_{\mathrm{MOM}} = \frac{\bar{X}_n}{\alpha}.$$

So the same CLT calculation gives

$$\sqrt{n}(\hat{\theta}_{\mathrm{MOM}} - \theta) \Rightarrow \mathcal{N}\left(0, \frac{\theta^2}{\alpha}\right).$$

ARE (your conclusion).

$$\mathrm{ARE}(\hat{\theta}_{\mathrm{MOM}}, \hat{\theta}_{\mathrm{MLE}}) = 1.$$

Solution 3

Normal density.

$$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Then

$$\mathbb{E}[X] = \int_{-\infty}^{+\infty} x f_X(x)\,dx = \frac{1}{\sqrt{2\pi}\,\sigma}\int_{-\infty}^{+\infty} x\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\,dx.$$

Change of variable (your substitution). Let $t = x - \mu$, so $x = t + \mu$ and $dx = dt$. Then

$$\mathbb{E}[X] = \frac{1}{\sqrt{2\pi}\,\sigma}\int_{-\infty}^{+\infty} (t+\mu)\exp\left(-\frac{t^2}{2\sigma^2}\right)\,dt.$$

Split:

$$\mathbb{E}[X] = \frac{1}{\sqrt{2\pi}\,\sigma}\left[\int_{-\infty}^{+\infty} t\,\exp\left(-\frac{t^2}{2\sigma^2}\right)\,dt + \mu\int_{-\infty}^{+\infty}\exp\left(-\frac{t^2}{2\sigma^2}\right)\,dt\right].$$

(1) First integral (your evaluation to 0). Define

$$I_1 := \int_{-\infty}^{+\infty} t \exp\left(-\frac{t^2}{2\sigma^2}\right) dt.$$

Use $z = -\frac{t^2}{2\sigma^2}$, so $dz = -\frac{t}{\sigma^2}dt$ and $t\,dt = -\sigma^2\,dz$. Formally,

$$I_1 = -\sigma^2 \int e^z\,dz = -\sigma^2 e^z\Big|_{-\infty}^{+\infty}.$$

To evaluate correctly over $(-\infty, +\infty)$, split:

$$I_1 = \left(\int_{-\infty}^{0} + \int_{0}^{+\infty}\right) t\, e^{-\frac{t^2}{2\sigma^2}}\,dt.$$

On $(0, +\infty)$:

$$\int_0^{+\infty} t\, e^{-\frac{t^2}{2\sigma^2}}\,dt = -\sigma^2 e^{-\frac{t^2}{2\sigma^2}}\Big|_0^{+\infty} = -\sigma^2(0-1) = \sigma^2.$$

On $(-\infty, 0)$:

$$\int_{-\infty}^0 t\, e^{-\frac{t^2}{2\sigma^2}}\,dt = -\sigma^2 e^{-\frac{t^2}{2\sigma^2}}\Big|_{-\infty}^{0} = -\sigma^2(1-0) = -\sigma^2.$$

Hence

$$I_1 = \sigma^2 - \sigma^2 = 0.$$

(2) Second integral (your normalization). Let

$$I_2 := \int_{-\infty}^{+\infty} \exp\left(-\frac{t^2}{2\sigma^2}\right) dt.$$

Since $\int_{-\infty}^{+\infty} f_X(x)\,dx = 1$, using the same substitution $t = x - \mu$,

$$\frac{1}{\sqrt{2\pi}\,\sigma} I_2 = 1 \quad \Longrightarrow \quad I_2 = \sqrt{2\pi}\,\sigma.$$

Conclusion (your final line).

$$\mathbb{E}[X] = \frac{1}{\sqrt{2\pi}\,\sigma}\big(I_1 + \mu I_2\big) = \frac{1}{\sqrt{2\pi}\,\sigma}\big(0 + \mu\sqrt{2\pi}\,\sigma\big) = \mu.$$

$$\boxed{\mathbb{E}[X] = \mu.}$$

Appendice

Archivio storico - testi disponibili

Le prove seguenti sono conservate integralmente. Questa appendice non e' una raccolta di prove svolte.

I fascicoli denominati NSMF trattano inferenza statistica (MLE, test, sufficienza), quindi sono raccolti qui.

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

January 17, 2011

Time allowed: two hours. PLEASE, make an effort to write in a legible and organized way.

Question 1. Introduce the likelihood function and describe the maximum likelihood method for constructing estimators. Provide (and solve) at least one problem in which the maximum likelihood method cannot be applied.

Question 2. The response time of a computer to a terminal's input is modeled by means of an Exponential random variable with an unknown parameter θ. In an experiment n response times $X_1, \ldots, X_n$ are measured. Prove that

$$\bar{T}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$$

is an unbiased estimator of $1/\theta$.[1] Is $\tilde{T}_n = 1/\bar{T}_n$ an unbiased estimator for the parameter θ? Derive an unbiased estimator S_n for the parameter θ and compute the corresponding MSE. Is S_n a consistent estimator for the parameter θ?

[1] **Hint**: the random variable $n\bar{T}_n$ has density function $f(x;\theta) = \frac{\theta^n}{(n-1)!}x^{n-1}e^{-\theta x}$.

date: January 16, 2011

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

February 14, 2011

Time allowed: two hours. PLEASE, make an effort to write in a legible and organized way.

Question 1. The sufficiency in the theory of statistical inference based on the likelihood function: definitions, main properties, theorems and examples.

Question 2. The weekly number of arrests for failure in a department is distributed according to a Poisson distribution with parameter λ, and the number of arrests in subsequent weeks are assumed to be independent. Let $X_1, \ldots, X_{10}$ be the number of arrests in 10 weeks and let

$$\bar{X} = \frac{1}{10} \sum_{i=1}^{10} X_i.$$

Under the above assumption

- determine the value of t such that $\{\bar{X} > t\}$ is a rejection region of level $\alpha = 0.05$ for testing hyphotesis $H_0 : \lambda = 2$ against $H_1 : \lambda = 3$. Which is the power of the test? How big should be t if $\bar{X}$ were the average number of failures over 20 weeks? What would be the power of the test?
- What would the rejection region be according to the Neyman Pearson approach based on the likelihood function?

date: February 13, 2011

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

January 31, 2011

Time allowed: two hours. PLEASE, make an effort to write in a legible and organized way.

Question 1. The Neyman–Pearson lemma in the theory of statistical hyphotesis testing. Provide and solve one example of application of the Neyman–Pearson lemma (or its generalizations) in statistical hyphotesis testing under Gaussian assumption for the random sample.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample distributed according to a random variable X such that

$$f_X(x;\theta) = \theta x^{\theta-1} \mathbb{1}_{(0,1)}(x).$$

Find the maximum likelihood estimator $\tilde{\theta}_n$ for the parameter θ. Which is a sufficient statistic for the parameter θ? Find the moment–based estimator $\bar{\theta}_n$ for the parameter θ. Check if the estimators $\tilde{\theta}_n$ and $\bar{\theta}_n$ are unbiased and consistent for the parameter θ.

date: January 27, 2011

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

December 19, 2011

Time allowed: one hour and a half. PLEASE, make an effort to write in a legible and organized way.

Question 1. Introduce the likelihood function and describe the maximum likelihood method applied to the theory of point estimation. Provide and solve at least one problem of point estimation in which the usual conditions of regularity for the likelihood function are not satisfied.

Question 2. In a biological population only the 40% of the individuals survive after the first week of life. The population census after one week detects k individuals. We want to estimate the number N of hatched eggs. Which is the maximum likelihood estimator $\hat{N}$ of N? Compute the value of of the estimator $\hat{N}$ for a number of individuals $k = 400$.

date: December 12, 2011

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

February 4, 2013

Time allowed: one hour and a half. PLEASE, make an effort to write in a legible and organized way.

Question 1. Describe the theory of hypothesis testing according to the Neyman and Pearson lemma. Let $(X_1, \ldots, X_n)$ be a sample from a random variable X distributed according to the density function

$$f_X(x;\theta) = \exp\{A(x)Q(\theta) + C(x) - K(\theta)\}\, \mathbb{1}_S(x)$$

with S being some subset of $\mathbb{R}$ and with A, Q, C, K being known functions. Let Q be a monotone increasing function, which is the shape of the rejection region for testing H_0 : $\theta = \theta_0$ versus $H_1 : \theta = \theta_1 > \theta_0$?

Question 2. Let $(X_1, \ldots, X_n)$ be a sample from a random variable X distributed according to the density function

$$f_X(x;\theta) = \frac{4x^2}{\theta^3\sqrt{\pi}} \mathrm{e}^{-x^2/\theta^2} \mathbb{1}_{\mathbb{R}^+}(x)$$

with $\theta > 0$. Find a sufficient statistics for θ and determine the maximum likelihood estimator $\hat{\theta}_{ML}$ for θ. Furthermore

- compute the second moment of $\hat{\theta}_{ML}$; can you infer that $\hat{\theta}_{ML}$ is an unbiased estimator for θ?
- compute the first moment of X_1; find an unbiased estimator for θ and compute its mean square error.

date: January 31, 2013

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

December 17, 2012

Time allowed: one hour and a half. PLEASE, make an effort to write in a legible and organized way.

Question 1. Describe the finite and asymptotic efficiency for an estimator T_n of a parameter θ. State and prove a necessary and sufficient condition for the existence of an efficient and unbiased estimator T_n of a parameter θ.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a random variable X distributed according to the following parametric family

$$f_X(x;\theta) = \exp\{-(x-\theta)\}\mathbb{1}_{(\theta,+\infty)}(x).$$

- Find the maximum likelihood estimator $\hat{\theta}_{ML}$ for the parameter θ. Check if the estimator $\hat{\theta}_{ML}$ is unbiased and (absolute) efficient for θ, for a finite sample size and asymptotically.
- Identify, if possible, the parametric family associated with the distribution of the estimator $\hat{\theta}_{ML}$.
- Provide an optimal rejection region of size α to test the following hypothesis: H_0 : $\theta = \theta_0$ versus $H_1 : \theta > \theta_0$.

date: December 16, 2012

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

February 4, 2013

Time allowed: one hour and a half. PLEASE, make an effort to write in a legible and organized way.

Question 1. Describe the theory of hypothesis testing according to the Neyman and Pearson lemma. Let $(X_1, \ldots, X_n)$ be a sample from a random variable X distributed according to the density function

$$f_X(x;\theta) = \exp\{A(x)Q(\theta) + C(x) - K(\theta)\}\, \mathbb{1}_S(x)$$

with S being some subset of $\mathbb{R}$ and with A, Q, C, K being known functions. Let Q be a monotone increasing function, which is the shape of the rejection region for testing H_0 : $\theta = \theta_0$ versus $H_1 : \theta = \theta_1 > \theta_0$?

Question 2. Let $(X_1, \ldots, X_n)$ be a sample from a random variable X distributed according to the density function

$$f_X(x;\theta) = \frac{4x^2}{\theta^3\sqrt{\pi}} \mathrm{e}^{-x^2/\theta^2} \mathbb{1}_{\mathbb{R}^+}(x)$$

with $\theta > 0$. Find a sufficient statistics for θ and determine the maximum likelihood estimator $\hat{\theta}_{ML}$ for θ. Furthermore

- compute the second moment of $\hat{\theta}_{ML}$; can you infer that $\hat{\theta}_{ML}$ is an unbiased estimator for θ?
- compute the first moment of X_1; find an unbiased estimator for θ and compute its mean square error.

date: January 31, 2013

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

June 10, 2013

Time allowed: one hour and a half. PLEASE, make an effort to write in a legible and organized way.

Question 1. Introduce the likelihood function and describe the maximum likelihood method applied to the theory of point estimation.

- Let $(X_1, \ldots, X_n)$ be a sample from a random variable X distributed according to the density function

$$f_X(x;\theta) = \frac{4x^2}{\theta^3\sqrt{\pi}} e^{-x^2/\theta^2} 1_{\mathbb{R}^+}(x)$$

with $\theta > 0$. Find a sufficient statistics for θ and determine the maximum likelihood estimator $\hat{\theta}_{ML}$ for θ.

- Provide and solve at least one problem of point estimation in which the usual conditions of regularity for the likelihood function are not satisfied.

Question 2. The central unit of a computer receives service requests from users. It is known that every service binds the memory for an exponential time with a known parmeter λ. If we know that the central unit was active for a time $X = x$, which is the maximum likelihood estimator for the number n of service requests? Suppose that $\lambda = 3$ and $x = 227$, which is the estimated value for n?

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

June 05, 2020

Time allowed: one hour and a half. PLEASE, make an effort to write in a legible and organized way.

Question 1. Illustrate the theory of statistical hyphotesis testing according to the Neyman and Pearson lemma and its likelihood ratio generalization. Let $(X_1, \ldots, X_n)$ be a random sample from a radom variable $X \sim N(\theta, \sigma^2)$ with σ^2 unknown. Describe the shape of the rejection region for testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

Question 2. A problem in surveys is that people tend to lie to protect their privacy. To overcome this drawback a common strategy is the following: we ask to the interviewee to flip a coin that produce "YES" with probability p and "NO" with probability $(1-p)$, and then refer if the outcome - which remains secret - coincides with his answer to the survey's question. It is assumed that the interviewee does not lie to this question. Suppose that the aim of the survey is to estimate the proportion θ of individuals in a certain group in the population:

- which is the probability that the interviewee's answer is "YES"?
- if Y_n is the number of interviewees that answered "YES", what is the distribution of Y_n? what is an unbiased estimator for θ? what is its variance? Is it a consistent estimator?

Question 3. Let $(\Omega, \mathscr{F}, \mathbb{P})$ be a probability space, and let A and B be events in $\mathscr{F}$ such that $\mathbb{P}(A) = 0.4$ and $\mathbb{P}(B) = 0.7$. Can A and B be mutually exclusive? If $\mathbb{P}(A) = 0.4$ and $\mathbb{P}(B) = 0.3$ can A and B be mutually exclusive? Motivate your answers.

date: May 28, 2020

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

December 21, 2020 - slot 1 -

Time allowed: one hour and 15 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Illustrate the theory of statistical hyphotesis testing according to the likelihood ratio test.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a random variable X distributed according to the following parametric family

$$f_X(x;\theta) = \exp\{-(x-\theta)\}\mathbb{1}_{(\theta,+\infty)}(x).$$

Find the maximum likelihood estimator $\hat{\theta}_{ML}$ for the parameter θ, and verify if the estimator is unbiased for a finite sample size and asymptotically. Find the optimal rejection region of size α to test the following hypothesis: $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$

Question 3. Let X be a random variable distributed according to a Poisson distribution with parameter $\lambda > 0$. Compute the mean and the variance of X.

date: December 9, 2020

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

January 21, 2021

Time allowed: one hour and 30 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Let $(X_1, \ldots, X_n)$ be a random sample from the Uniform distribution on the set $(0, \theta)$, with $\theta > 0$. Show that

i)
$$\frac{d}{d\theta} \int x f_\theta(x) dx \neq \int x \frac{d}{d\theta} f_\theta(x) dx,$$
where f_θ is the density function of the largest order statistics $X_{(n)}$.

ii) Show that the Cramer-Rao lower bound does not hold for the estimator $\hat{\theta} = (n+1)X_{(n)}/n$ of θ.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a Gaussian distribution with parameter $(\mu, 1)$, where μ is unknown. Instead of recording all the observations, one records only whether the observation is less than 0. Find the maximum likelihood estimator of μ based on the recorded information, and discuss its properties.

date: January 5, 2021

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

January 22, 2021 - slot 1 -

Time allowed: one hour and 15 minutes . PLEASE, make an effort to write in a legible and organized way.

Question 1. Introduce the Fisher information and discuss its role within the problem of point estimation.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a random variable X with density function $f_X(x;\theta) = 2^{-1}\mathrm{e}^{-|x-\theta|}\mathbb{1}_{\mathbb{R}}(x)$, and where $\theta \in \mathbb{R}$. Within this parametric model:

- study sufficiency properties of $f_X(x;\theta)$;
- find an estimator of θ with the method of moments;
- find the maximum likelihood estimator of θ;
- does $f_X(x;\theta)$ belong to the exponential family?

Question 3. A process for putting chocolate chips into cookies is random and the number of choc chips in a cookie has a Poisson distribution with mean λ. Find an expression for the probability that a cookie contains less than 3 choc chips.

date: January 5, 2021

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

January 22, 2021 - slot 2 -

Time allowed: 1 hour and 15 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Describe the property of efficiency, finite and asymptotic, in the theory of point estimation based on the likelihood function.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a random variable X with density function $f_\theta(x) = \theta(1+x)^{-(1+\theta)}\mathbb{1}_{\mathbb{R}^+}(x)$, and where $\theta \in \mathbb{R}^+$. By using this parametric model:

- find an estimator $T_{1,n}$ of θ with the method of moments;
- find the maximum likelihood estimator $T_{2,n}$ of θ^{-1};
- are $T_{1,n}$ and $T_{2,n}$ unbiased?

Question 3. Three babies are give a weekly health check at a clinic, and then returned randomly to their mothers. What is the probability that at least one baby goes to the right mother?

date: January 5, 2021

QUANTITATIVE METHODS FOR ECONOMICS

February 08, 2021 - slot 1 -

Time allowed: 1 hour and 15 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Consider a parametric family $F = \{f_\theta : \theta \in \Theta\}$ and describe the properties of unbiasedness and efficiency for an estimator T of θ.

Question 2. Let $f_X(x;\beta) = \beta^{-1}\mathrm{e}^{-x/\beta}\mathbb{1}_{(0,+\infty)}(x)$, and let $(X_1,\ldots,X_n)$ be a random sample from the statistical model $\{f_X(x;\beta) : \beta > 0\}$. Find the maximum likelihood estimator of the parameter β under the assumption that the sole information available from the sample is that $0 \leq k \leq n$ observations are less or equal than M, where M is a fixed positive number.

Question 3. An urn contains r red balls and b blue balls, $r \geq 1$, $b \geq 3$. Three balls are selected, without replacement, from the urn. Using the notion of conditional probability to simplify the problem, find the probability of the sequence Blue, Red, Blue.

date: February 1, 2021

NUMERICAL AND STATISTICAL METHODS FOR FINANCE

September 9th, 2021

Time allowed: one hour and 15 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Let $(X_1, \ldots, X_n)$ be a random sample from a regular parametric family $f_X(x;\theta)$. Describe the property of efficiency, finite and asymptotic, for an estimator T_n of θ. State and prove a necessary and sufficient condition for the existence of an efficient and unbiased estimator T_n of θ.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a random variable X distributed according to the following parametric family

$$f_X(x;\theta) = \exp\{-(x-\theta)\}\mathbb{1}_{(\theta,+\infty)}(x).$$

- Find the maximum likelihood estimator $\hat{\theta}_{ML}$ for the parameter θ. Check if the ML estimator $\hat{\theta}_{ML}$ is unbiased for θ, for a finite sample size and asymptotically, and if it is consistent for θ.
- Provide an optimal rejection region of size α to test the following hypothesis: $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1 > \theta_0$.

Question 3. A fair coin is thrown repeatedly. What is the probability that on the nth throw: i) a head appears for the first time; ii) the number of heads and tails to date are equal?

date: September 7, 2021

February 2, 2017

Time allowed: two hours and a half. PLEASE, make an effort to write in a legible and organized way.

Question 1. Let $\mathbf{X} = (X_1, \ldots, X_n)$ be a random sample from a parametric family $\{f_\theta(x) : \theta \in \Theta\}$, with $\Theta \subset \mathbb{R}$.

i) Describe the property of efficiency for an estimator T of θ. State and prove a necessary and sufficient condition for the existence of an efficient and unbiased estimator T of θ.

ii) Let ψ defined on Θ, and let T be an unbiased estimator of $\psi(\theta)$ with $\mathbb{E}[T^2] < +\infty$ for all $\theta \in \Theta$. If $\theta \neq \varphi$, assume that f_θ and f_φ are different ad assume that there exists a $\varphi \in \Theta$ such that $\theta \neq \varphi$ and $S(\theta) = \{f_\theta(\mathbf{x}) > 0\} \supset S(\varphi) = f_\varphi(\mathbf{x}) > 0$. Show that

$$Var(T(\mathbf{X})) \geq \sup_{\{\varphi : S(\varphi) \subset S(\theta), \varphi \neq \theta\}} \frac{[\psi(\varphi) - \psi(\theta)]^2}{Var\left(\frac{f_\varphi(\mathbf{X})}{f_\theta(\mathbf{X})}\right)}$$

Question 2.

$\theta > 0$

Let $\mathbf{X} = (X_1, \ldots, X_n)$ be a random sample from a parametric family $\{f_\theta(\mathbf{x}) : \theta \in \Theta\}$, where

$$f_\theta(x) = \frac{1}{\theta} x^{\frac{1-\theta}{\theta}} \mathbb{1}_{(0,1)}(x);$$

i) find, if this exists, the UMP test of size α for testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$;

ii) for a sample size $n = 2$, $\theta_0 = 1$, $\alpha = 0.05$ represents the power function of the UMP test.

date: February 1, 2017

STATISTICAL INFERENCE

February 20th, 2019

Time allowed: two hours and 15 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Let $\mathbf{X} = (X_1, \ldots, X_n)$ be a random sample from a parametric family $\{f_\theta(x) : \theta \in \Theta\}$, with $\Theta \subset \mathbb{R}$.

i) Describe the property of efficiency for an estimator T of θ. State and prove a necessary and sufficient condition for the existence of an efficient and unbiased estimator T of θ.

ii) Prove that the exponential family of distributions is a parametric family of distributions for which it exists an efficient estimator (biased or unbiased) for the parameter of interest.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a parametric family $\{f_\theta : \theta \in \Theta\}$ such that

$$f_\theta(x) = \begin{cases} \frac{2x}{\alpha\theta} & \text{if } 0 \leq x \leq \theta, \\ \frac{2(\alpha - x)}{\alpha(\alpha - \theta)} & \text{if } \theta \leq x \leq \alpha, \\ 0 & \text{otherwise} \end{cases}$$

where $\alpha > 0$ is known. What is the maximum likelihood estimator of θ? Compute the maximum likelihood estimator using the following setting: $n = 3$, $(x_1, x_2, x_3) = (1, 2, 4)$ and $\alpha = 5$.

date: February 7, 2019

STATISTICAL INFERENCE

February 24th, 2019

Time allowed: two hours and a 15 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Illustrate the theory of statistical hyphotesis testing according to the approach of the likelihood ratio test. Let $\bar{X} := (X_1, \ldots, X_n)$ be random sample from $X \sim N(\mu_x, \sigma_x^2)$ and let $\bar{Y} := (Y_1, \ldots, Y_m)$ be random sample from $Y \sim N(\mu_y, \sigma_y^2)$, independent of $\bar{X}$. Describe the shape and distribution of the rejection region for testing the hyphotesis

$$H_0 : \sigma_x^2 = \sigma_y^2 \qquad H_1 : \sigma_x^2 \neq \sigma_y^2.$$

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a Poisson random variable with parameter $\lambda > 0$, and consider the problem of estimating the functional $\theta = \mathrm{e}^{-\lambda}$. For $n = 1$, show that $T_1 = \delta_{X_1}$ is the unique unbiased estimator with minimum variance for θ. For an arbitrary sample size $n \geq 1$, is $T_n = n^{-1} \sum_{i=1}^{n} \delta_{X_i}$ an unbiased and efficient estimator for θ?

date: February 24, 2019

STATISTICAL INFERENCE

June 11th, 2019

Time allowed: two hours and a 15 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Let $Y = X\beta + \varepsilon$, where: i) $Y = (Y_1, \ldots, Y_n)^T$ is the response variable; ii) $X = (x_{ij})_{1 \le i \le n, 1 \le j \le p}$ is the design matrix with $n \ge p$; iii) $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_n)^T$ is the error component; iv) $\beta = (\beta_1, \ldots, \beta_p)^T$ is the parameter. Describe the use of the least square criterion for estimating β, and show that estimates obtained with the least square criterion coincide with MLE estimates obtained under a Gaussian assumption for ε. Consider the following test

$$\begin{cases} H_0 & : \beta_r = 0, \\ H_1 & : \beta_r \neq 0, \end{cases} \tag{0.1}$$

for $1 \le r \le p$. Using the LR approach, propose a statistic for testing (0.1) and determine its distribution.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a Poisson random variable X with (mean) parameter $\lambda > 0$, and consider the problem of estimating the functional $\theta = e^{-\lambda}$. Let

$$\delta_X = \begin{cases} 1 & \text{if } X = 0, \\ 0 & \text{if } X \ge 1. \end{cases}$$

For $n = 1$, show that $T_1 = \delta_{X_1}$ is the unique unbiased estimator with minimum variance for θ. For an arbitrary sample size $n \ge 1$, is $T_n = n^{-1} \sum_{1 \le i \le n} \delta_{X_i}$ an unbiased and efficient estimator for θ?

date: June 7, 2019

STATISTICAL INFERENCE

Written Exam
January 2020

Exercise 1

Let $X = (X_1, ...X_n)$ be a random sample from a parametric family $\{f_\theta(x) : \theta \in \Theta \subset \mathbb{R}\}$

(i) Define the property of Efficiency (both asymptotical and non-asymptotical)
(ii) Look at February 2017 Ex.1.(ii)

Exercise 2

Let $X = (X_1, ...X_n)$ be a random sample from a parametric family $\{f_\mu(x) : \mu \in M \subset \mathbb{R}\}$, s.t. $\mathbb{E}[X_1] = \mu$, $Var(X_1) = 1$ and $\mathbb{E}X_1^4 < +\infty$. Let T_1 and T_2 the following unbiased estimators for μ^2:

$$T_{1,n}(X) = \frac{1}{n}\sum X_i^2 - 1 \quad T_{2,n}(X) = (\bar{X}_n)^2 - \frac{1}{n}$$

(i) Find the relative asymptotic efficiency of T_1 and T_2, i.e. $lim_{n\longrightarrow\infty} eff(T_{1,n}, T_{2,n})$
(ii) Show that, if $X_i - \mu$ is symmetric about 0, then $lim_{n\longrightarrow\infty} eff(T_{1,n}, T_{2,n}) \leq 1$
(iii) Find a parametric family s.t. $lim_{n\longrightarrow\infty} eff(T_{1,n}, T_{2,n}) > 1$

Hint 1: if $\sqrt{n}(X_n - \theta) \longrightarrow N(0, \sigma^2)$, then $\sqrt{n}(g(X_n) - g(\theta)) \longrightarrow N(0, g'(\theta)^2\sigma^2)$, $\forall g$ s.t. $\exists g'(\theta) \neq 0$
Hint 2: if $X - \mu$ is sym. about 0, then $\mathbb{E}[(X - \mu)^r] = 0$, for any r odd integer.
Definition: the relative efficiency of two estimators is defined as:

$$eff(T_1, T_2) = \frac{\mathbb{E}[(T_2 - \theta)^2]}{\mathbb{E}[(T_1 - \theta)^2]}$$

1) Speak about LR test for general hypothesis
(case simple-c and in assymptotic theory)

2) Show that, for a parametric family $\mathcal{F} = \{f_\theta(x) : \theta \in \Theta\}$
to have the monotone likelihood ratio is equivalent to satisfy
one of the following

i) $\frac{d^2}{d\theta dx} \log f_\theta(x) \geq 0$

ii) $f_\theta(x) \frac{d^2}{d\theta dx} f_\theta(x) \geq \frac{df_\theta(x)}{d\theta} \frac{df_\theta(x)}{dx}$

3) Let p be a parameter family of densities from a
non-central (parameter of non centrality $\theta \geq 0$) chi square
with one d.o.f. Show that it has the monotone lr.

4) $\{f(x;\theta) : \theta \in \Theta\}$ with $f(x;\theta) = \theta x^{-2} \prod_{[\theta,+\infty[}(x)$
FIND an UMP SIZE-α TEST FOR
$H_0: \theta = \theta_0$ vs $H_0: \theta = \theta_1$

STATISTICAL INFERENCE

February 09, 2021

Time allowed: one hour and 30 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Let X_{ij}, for $j = 1, \ldots, r > 1$ and $i = 1, \ldots, n$ be a collection of random variables independently distributed according to Gaussian distributions with parameters (μ_i, σ^2). Find the maximum likelihood estimator of $\theta = (\mu_1, \ldots, \mu_n, \sigma^2)$, and verify consistency properties (as $n \to +\infty$) of the maximum likelihood estimator of σ^2.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a parametric family $\{f_\theta(x) : \theta \in \Theta\}$ on $\mathcal{X} \subset \mathbb{R}$, with $\Theta = \mathbb{R}^+$. Let θ_0 be a positive constant. Find a UMP test of size α for testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$ under the following assumptions

i)

$$f_\theta(x) = \theta^{-1} \mathrm{e}^{-x/\theta} \mathbb{1}_{\mathbb{R}^+}(x)$$

ii)

$$f_\theta(x) = \theta^{-1} x^{\theta - 1} \mathbb{1}_{(0,1)}(x)$$

iii)

$$f_\theta(x) = \theta^{-4} 4x^3 \mathrm{e}^{-(x/\theta)^4} \mathbb{1}_{\mathbb{R}^+}(x)$$

date: January 29, 2021

iv)

$$f_\theta(x) = (2\pi\theta)^{-1/2} \mathrm{e}^{-(x-1)^2/\theta} \mathbb{1}_{\mathbb{R}}(x)$$

STATISTICAL INFERENCE

July 09, 2021

Time allowed: one hours and 30 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Let $X = (X_1, \ldots, X_n)$ be a random sample from the discrete Uniform distribution on the set $\{1, \ldots, \theta\}$ for $\theta \in \mathbb{N}$, and let $X_{(n)} = \max\{X_1, \ldots, X_n\}$ be the sample maximum.

i) Consider the problem of testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$, where $\theta_0 > 0$ is known. Show that

$$T_*(X) = \begin{cases} 1 & \text{if } X_{(n)} > \theta_0 \\ \\ \alpha & \text{if } X_{(n)} \leq \theta_0 \end{cases}$$

is a uniformly most powerful test of size α.

ii) Consider the problem of testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$ where $\theta_0 > 0$ is known. Show that

$$T_*(X) = \begin{cases} 1 & \text{if } X_{(n)} > \theta_0 \text{ or } X_{(n)} \leq \theta_0 \alpha^{1/n} \\ \\ 0 & \text{otherwise} \end{cases}$$

is a uniformly most powerful test of size α.

iii) Show that the results displayed in i) and ii) still hold if the discrete uniform distribution is replaced by the continuous uniform distribution on the interval $(0, \theta)$, where $\theta > 0$.

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a Gaussian distribution with

parameter $(\mu, 1)$. Instead of recording observations, one records only whether observations are less than 0. Find the maximum likelihood estimator of μ based on the recorded information.

STATISTICAL INFERENCE

June 23, 2021

Time allowed: one hours and 30 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Let $(X_1, \ldots, X_n)$ be a random sample from the Uniform distribution on $(\theta, \theta + |\theta|)$. Find the maximum likelihood estimator of θ when: i) $\theta \in (0, +\infty)$; ii) $\theta \in (-\infty, 0)$; iii) $\theta \in \mathbb{R} - \{0\}$.

Question 2. Let $((Y_1, Z_1), \ldots, (Y_n, Z_n))$ be a random sample from (Y, Z), where (Y, Z) is a pair of independent random variable with Y and Y being independent Exponential random variables with scale parameters $\lambda > 0$ and $\mu > 0$, respectively. Consider the following questions:

i) find the maximum likelihood estimator of (λ, μ);

ii) suppose that we observe (X_i, Δ_i), where: i)$X_i = \min\{Y_i, Z_i\}$; ii) $\Delta_i = 1$ if $X_i = Y_i$ and $\Delta_i = 0$ if $X_i = Z_i$. Find the maximum likelihood estimator of (λ, μ).

Question 3. Let X be a sample from a population P, and let P_0 and P_1 be two known populations. Suppose that T_* is UMP test of size $\alpha \in (0, 1)$ for testing $H_0 : P = P_0$ versus $H_1 : P = P_1$ and that $\beta < 1$, where β is the power of T_* when H_1 is true. Show that $1 - T_*$ is a UMP test of size $1 - \beta$ for testing $H_0 : P = P_1$ versus $H_1 : P = P_0$.

date: June 18, 2021

1 $(X_1, \dots, X_n)$ gaussian with parametres $(\mu, 1)$ truncated in (α, β), $\alpha < \beta$. Show that the sample mean is the unique maximum likelihood estimator for $\theta = E[X]$. Then, show that it is asymptotically efficient (or something like that, some properties when $n \to \infty$)

2 $(Y_1, \dots, Y_n) \sim U(0, \theta)$, $\theta \in (1, +\infty)$. The only information we have from the sample is:

$$X_i = Y_i \mathbb{1}_{[1, +\infty)}(Y_i) + \mathbb{1}_{(0,1)}(Y_i)$$

Find a UMP test of size $\alpha \in (0,1)$ for testing $H_0 : \theta \leq \theta_0$ versus $H_1 : \theta > \theta_0$.

STATISTICAL INFERENCE

June 15, 2022

Time allowed: one hour and 30 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Consider sample of size n to be modeled as a collection of $(Y_1, \ldots, Y_n)$ independent random variables such that Y_i is distributed as a Bernoulli distribution with parameter

$$w_i = \frac{\mathrm{e}^{\alpha+\beta x_i}}{1+\mathrm{e}^{\alpha+\beta x_i}},$$

where $(x_1, \ldots, x_n)$ is a collection of known constants, i.e. no random, and (α, β) is a pair of unknown parameter. Provides maximum likelihood estimators for the parameter (α, β).

Question 2. Let $(X_1, \ldots, X_n)$ be a random sample from a parametric family $\{f_\theta(x) : \theta \in \Theta\}$ on $\mathcal{X} \subset \mathbb{R}$, with $\Theta \subset \mathbb{R}$. Find a UMP test of size α for testing the hypothesis $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1$ under the modeling assumption $f_\theta(x) = \theta x^{-2} \mathbb{1}_{\{\theta, +\infty\}}(x)$.

date: June 1, 2022

STATISTICAL INFERENCE

July 4, 2022

Time allowed: one hour and 30 minutes. PLEASE, make an effort to write in a legible and organized way.

Question 1. Let $(X_1, \ldots, X_n)$ be a random sample from a random variable X whose distribution has density function f_θ, for some $\theta \in \Theta$. Find the MLE of θ in these cases:

i) f_θ is the Bernoulli probability mass function with parameter θ, $\theta \in [1/2, 3/4]$;

ii) f_θ is the Gaussian density function with parameter (θ, θ^2), $\theta \in \mathbb{R}$ and $\theta \neq 0$.

Question 2. Let F and G be two known cumulative distribution functions on the real line $\mathbb{R}$, and consider a single observation modeled as a random variable X distributed according to the cumulative distribution function $\theta F + (1-\theta)G$ where $\theta \in [0,1]$ is unknown. Consider the problem of testing $H_0 : \theta \in [\theta_1, \theta_2]$ versus $H_1 : \theta \notin [\theta_1, \theta_2]$, where $0 < \theta_1 \leq \theta_2 < 1$ are given constants. Show that a UMP test does not exist.

date: June 22, 2022

e STINF (Exam 2023-07-06)

Question 1. Let $(X_1, \ldots, X_n)$ be a random sample from a distribution with density function $f_X(x;\theta) = \sigma^{-1}\exp(-(x-\mu)/\sigma)/(1+\exp(-(x-\mu)/\sigma))^2$ with $\mu \in \mathbb{R}$ and $\sigma > 0$. Apply one step of the Netwon-Rapson and Fisher scoring methods to find:

- the maximum likelihood estimator of μ when σ is known
- the maximum likelihood estimator of σ when μ is known

Question 2. Let $\bar{X} = (X_1, \ldots, X_n)$ be a random sample from $N \sim (\mu_x, \sigma^2)$ and $\bar{Y} = (Y_1, \ldots, Y_m)$ be a random sample from $N \sim (\mu_y, \sigma^2)$. Assuming $\bar{X}$ is independent of $\bar{Y}$, describe the shape of the rejection region for testing the hypotheses:

$$H_0 : \mu_x - \mu_y = 0 \qquad \text{vs} \qquad H_1 : \mu_x - \mu_y \neq 0$$

Sezione B: Probability Distributions (Esercizi di Probabilità di Base)

Questi esercizi riguardano il calcolo di probabilità su eventi, urne o proprietà di base delle distribuzioni discrete, senza richiedere inferenza statistica complessa.

- **Exam: June 05, 2020 – Question 3**
  - *Argomento:* Spazi di probabilità, eventi mutuamente esclusivi.
  - *Testo:* "Let (Ω, F, P) be a probability space... Can A and B be mutually exclusive?"
- **Exam: December 21, 2020 – Question 3**
  - *Argomento:* Distribuzione di Poisson (Media e Varianza).
  - *Testo:* "Let X be a random variable distributed according to a Poisson distribution... Compute the mean and the variance."
- **Exam: January 22, 2021 (Slot 1) – Question 3**
  - *Argomento:* Calcolo probabilità Poisson (cookie chips).
  - *Testo:* "A process for putting chocolate chips into cookies is random... Find an expression for the probability that a cookie contains less than 3 choc chips."
- **Exam: January 22, 2021 (Slot 2) – Question 3**
  - *Argomento:* Probabilità classica (Matching problem / Babies returned to mothers).
  - *Testo:* "Three babies are give a weekly health check... What is the probability that at least one baby goes to the right mother?"
- **Exam: February 08, 2021 (Slot 1) – Question 3**
  - *Argomento:* Probabilità condizionata ed estrazione da urne (senza reimmissione).
  - *Testo:* "An urn contains r red balls and b blue balls... find the probability of the sequence Blue, Red, Blue."
- **Exam: September 9, 2021 – Question 3**
  - *Argomento:* Lancio di monete (Primo successo, uguaglianza teste/croci).
  - *Testo:* "A fair coin is thrown repeatedly. What is the probability that on the n-th throw: i) a head appears for the first time..."

Sezione E: Likelihood and Maximum Likelihood Estimation (MLE)

Questi esercizi si concentrano sulla definizione di verosimiglianza, sul calcolo dello stimatore di massima verosimiglianza ($\hat{\theta}_{ML}$) e su casi in cui le condizioni di regolarità potrebbero non essere soddisfatte.

- **Exam: January 17, 2011 – Question 1**
  - *Argomento:* Teoria MLE e casi in cui non è applicabile.
  - *Testo:* "Introduce the likelihood function and describe the maximum likelihood method... Provide (and solve) at least one problem in which the maximum likelihood method cannot be applied."
- **Exam: December 19, 2011 – Question 1**
  - *Argomento:* Teoria MLE e regolarità.
  - *Testo:* "Describe the maximum likelihood method... Provide and solve at least one problem... in which the usual conditions of regularity... are not satisfied."
- **Exam: December 19, 2011 – Question 2**
  - *Argomento:* Stima di una popolazione (Uova schiuse) tramite MLE.
  - *Testo:* "We want to estimate the number N of hatched eggs. Which is the maximum likelihood estimator $\hat{N}$ of N?"
- **Exam: June 10, 2013 – Question 1**
  - *Argomento:* Teoria MLE e applicazione a densità specifica.
  - *Testo:* "Introduce the likelihood function... Let $(X_1, \ldots, X_n)$ be a sample... density $f_X(x;\theta) = \frac{4x^2}{\theta^3\sqrt{\pi}} e^{-x^2/\theta^2}$... determine the maximum likelihood estimator."

- **Exam: June 10, 2013 – Question 2**
  - *Argomento:* Esponenziale, stima di n dati $\sum x_i$.
  - *Testo:* "If we know that the central unit was active for a time $X = x$, which is the maximum likelihood estimator for the number n of service requests?"
- **Exam: January 21, 2021 – Question 2**
  - *Argomento:* MLE con dati censurati (solo indicatore < 0 registrato).
  - *Testo:* "Instead of recording all the observations, one records only whether the observation is less than 0. Find the maximum likelihood estimator of μ..."
- **Exam: February 08, 2021 (Slot 1) – Question 2**
  - *Argomento:* MLE con informazione parziale (conteggio osservazioni $\leq M$).
  - *Testo:* "Find the maximum likelihood estimator of the parameter β under the assumption that the sole information available... is that $0 \leq k \leq n$ observations are less or equal than M."

Sezione H: Sufficiency and Factorization

Esercizi focalizzati sul trovare statistiche sufficienti, spesso usando il Teorema di Fattorizzazione.

- **Exam: February 14, 2011 – Question 1**
  - *Argomento:* Teoria della sufficienza.
  - *Testo:* "The sufficiency in the theory of statistical inference based on the likelihood function: definitions, main properties, theorems and examples."
- **Exam: January 31, 2011 – Question 2 (Parte)**
  - *Argomento:* Statistica sufficiente per famiglia potenza.
  - *Testo:* "Which is a sufficient statistic for the parameter θ?" (Density $f_X(x;\theta) = \theta x^{\theta-1}$.)
- **Exam: February 4, 2013 – Question 2 (Parte)**
  - *Argomento:* Statistica sufficiente per densità specifica.
  - *Testo:* "Find a sufficient statistics for θ..." (Density $f_X(x;\theta) \propto x^2 e^{-x^2/\theta^2}$.)

Sezione J: Properties of Estimators (Unbiasedness, Consistency)

Esercizi che richiedono di verificare correttezza (unbiasedness), calcolare l'errore quadratico medio (MSE), verificare la consistenza o usare il Metodo dei Momenti.

- **Exam: January 17, 2011 – Question 2**
  - *Argomento:* Stimatori non distorti (inverso della media campionaria per esponenziale).
  - *Testo:* "Prove that $\bar{T}_n$ is an unbiased estimator of $1/\theta$. Is $1/\bar{T}_n$ an unbiased estimator for the parameter θ?... Is S_n a consistent estimator?"
- **Exam: January 31, 2011 – Question 2 (Parte)**
  - *Argomento:* Metodo dei momenti vs MLE, consistenza.
  - *Testo:* "Find the moment-based estimator $\bar{\theta}_n$... Check if the estimators $\tilde{\theta}_n$ and $\bar{\theta}_n$ are unbiased and consistent."
- **Exam: February 4, 2013 – Question 2 (Parte)**
  - *Argomento:* Secondo momento dello stimatore, correttezza asintotica.
  - *Testo:* "Compute the second moment of $\hat{\theta}_{ML}$; can you infer that $\hat{\theta}_{ML}$ is an unbiased estimator for θ?"
- **Exam: June 05, 2020 – Question 2**
  - *Argomento:* Randomized Response (Privacy), varianza e consistenza.
  - *Testo:* "What is an unbiased estimator for θ? What is its variance? Is it a consistent estimator?"
- **Exam: January 22, 2021 (Slot 1) – Question 2**

- *Argomento:* Famiglia esponenziale (Laplace), Momenti vs MLE.
- *Testo:* "Find an estimator of θ with the method of moments... does $f_X(x;\theta)$ belong to the exponential family?"

- **Exam: January 22, 2021 (Slot 2) – Question 2**
  - *Argomento:* Modello Pareto-simile, stima di θ^{-1}.
  - *Testo:* "Find an estimator $T_{1,n}$ of θ with the method of moments; find the maximum likelihood estimator $T_{2,n}$ of θ^{-1}; are $T_{1,n}$ and $T_{2,n}$ unbiased?"

Sezione K: Cramér–Rao Bound and Efficient Estimation

Esercizi teorici sull'efficienza, l'informazione di Fisher e controesempi (casi non regolari).

- **Exam: December 17, 2012 – Question 1**
  - *Argomento:* Efficienza asintotica e finita.
  - *Testo:* "Describe the finite and asymptotic efficiency... State and prove a necessary and sufficient condition for the existence of an efficient and unbiased estimator."

- **Exam: January 21, 2021 – Question 1**
  - *Argomento:* Uniforme $(0,\theta)$, fallimento regolarità (derivata sotto integrale) e CRLB.
  - *Testo:* "Show that $\frac{d}{d\theta}\int x f_\theta(x)\,dx \neq \int x \frac{d}{d\theta} f_\theta(x)\,dx$... Show that the Cramer-Rao lower bound does not hold..."

- **Exam: January 22, 2021 (Slot 1) – Question 1**
  - *Argomento:* Informazione di Fisher.
  - *Testo:* "Introduce the Fisher information and discuss its role within the problem of point estimation."

- **Exam: January 22, 2021 (Slot 2) – Question 1**
  - *Argomento:* Efficienza basata sulla funzione di verosimiglianza.
  - *Testo:* "Describe the property of efficiency, finite and asymptotic, in the theory of point estimation based on the likelihood function."

- **Exam: February 08, 2021 (Slot 1) – Question 1**
  - *Argomento:* Famiglia parametrica generale, correttezza ed efficienza.
  - *Testo:* "Consider a parametric family... describe the properties of unbiasedness and efficiency for an estimator T of θ."

- **Exam: September 9, 2021 – Question 1**
  - *Argomento:* Condizione necessaria e sufficiente per efficienza.
  - *Testo:* "State and prove a necessary and sufficient condition for the existence of an efficient and unbiased estimator T_n of θ."

Sezioni P, Q, R: Hypothesis Testing (Neyman–Pearson & LRT)

Esercizi sui Test d'Ipotesi, Lemma di Neyman–Pearson, Rapporto di Verosimiglianza e costruzione delle regioni di rifiuto.

- **Exam: January 31, 2011 – Question 1**
  - *Argomento:* Lemma di Neyman–Pearson (Teoria ed esempio Gaussiano).
  - *Testo:* "The Neyman–Pearson lemma... Provide and solve one example... under Gaussian assumption."

- **Exam: February 14, 2011 – Question 2**
  - *Argomento:* Poisson, Regione di rifiuto e Potenza.
  - *Testo:* "Determine the value of t such that $\{\bar{X} > t\}$ is a rejection region... Which is the power of the test?... according to the Neyman Pearson approach."

- **Exam: December 17, 2012 – Question 2**
  - *Argomento:* Esponenziale traslata (shift), MLE e Test d'ipotesi.

  - *Testo:* "Provide an optimal rejection region of size α to test... $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$."

- **Exam: February 4, 2013 – Question 1**
  - *Argomento:* Teoria NP e forma della regione di rifiuto per Famiglia Esponenziale.
  - *Testo:* "Describe the theory... Let Q be a monotone increasing function, which is the shape of the rejection region...?"

- **Exam: June 05, 2020 – Question 1**
  - *Argomento:* Teoria NP e LRT generalizzato. Caso Normale con varianza ignota.
  - *Testo:* "Illustrate the theory... its likelihood ratio generalization... Describe the shape of the rejection region for testing... (Normal with σ^2 unknown)."

- **Exam: December 21, 2020 – Question 1**
  - *Argomento:* Likelihood Ratio Test (Teoria).
  - *Testo:* "Illustrate the theory of statistical hyphotesis testing according to the likelihood ratio test."

- **Exam: December 21, 2020 – Question 2**
  - *Argomento:* Esponenziale traslata (Shift), MLE e Test.
  - *Testo:* "Find the optimal rejection region of size α to test... $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$."

- **Exam: September 9, 2021 – Question 2**
  - *Argomento:* Esponenziale traslata, MLE, Correttezza, Test d'ipotesi.
  - *Testo:* "Find the maximum likelihood estimator... Check if... unbiased... Provide an optimal rejection region..."

Exam: January 31, 2011

Question 1

The Neyman–Pearson lemma in the theory of statistical hyphotesis testing. Provide and solve one example of application of the Neyman–Pearson lemma (or its generalizations) in statistical hyphotesis testing under Gaussian assumption for the random sample.

Exam: February 14, 2011

Question 2

The weekly number of arrests for failure in a department is distributed according to a Poisson distribution with parameter λ, and the number of arrests in subsequent weeks are assumed to be independent. Let $X_1, \ldots, X_{10}$ be the number of arrests in 10 weeks and let

$$\bar{X} = \frac{1}{10} \sum_{i=1}^{10} X_i.$$

Under the above assumption

- determine the value of t such that $\{\bar{X} > t\}$ is a rejection region of level $\alpha = 0.05$ for testing hyphotesis $H_0 : \lambda = 2$ against $H_1 : \lambda = 3$. Which is the power of the test? How big should be t if $\bar{X}$ were the average number of failures over 20 weeks? What would be the power of the test?
- What would the rejection region be according to the Neyman–Pearson approach based on the likelihood function?

Exam: December 17, 2012

Question 2

Let $(X_1, \ldots, X_n)$ be a random sample from a random variable X distributed according to the following parametric family

$$f_X(x; \theta) = \exp\{-(x - \theta)\}\, 1_{(\theta, +\infty)}(x).$$

[...]

- Provide an optimal rejection region of size α to test the following hypothesis: $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$.

Exam: February 4, 2013

Question 1

Describe the theory of hypothesis testing according to the Neyman and Pearson lemma. Let $(X_1, \ldots, X_n)$ be a sample from a random variable X distributed according to the density function

$$f_X(x; \theta) = \exp\{A(x)Q(\theta) + C(x) - K(\theta)\}\, 1_S(x)$$

with S being some subset of R and with A, Q, C, K being known functions. Let Q be a monotone increasing function, which is the shape of the rejection region for testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1 > \theta_0$?

Exam: June 05, 2020

Question 1

Illustrate the theory of statistical hyphotesis testing according to the Neyman and Pearson lemma and its likelihood ratio generalization. Let $(X_1, \ldots, X_n)$ be a random sample from a radom variable $X \sim N(\theta, \sigma^2)$ with σ^2 unknown. Describe the shape of the rejection region for testing $H_0 : \theta = \theta_0$ versus $H_1 : \theta \neq \theta_0$.

Exam: December 21, 2020

Question 1

Illustrate the theory of statistical hyphotesis testing according to the likelihood ratio test.

Question 2

Let $(X_1, \ldots, X_n)$ be a random sample from a random variable X distributed according to the following parametric family

$$f_X(x;\theta) = \exp\{-(x-\theta)\}\, 1_{(\theta,+\infty)}(x).$$

[...] Find the optimal rejection region of size α to test the following hypothesis: $H_0 : \theta = \theta_0$ versus $H_1 : \theta > \theta_0$.

Exam: September 9, 2021

Question 2

Let $(X_1, \ldots, X_n)$ be a random sample from a random variable X distributed according to the following parametric family

$$f_X(x;\theta) = \exp\{-(x-\theta)\}\, 1_{(\theta,+\infty)}(x).$$

[...]

- Provide an optimal rejection region of size α to test the following hypothesis: $H_0 : \theta = \theta_0$ versus $H_1 : \theta = \theta_1 > \theta_0$.

Parte II

Prove generate: nessun fascicolo trovato

Nessuna prova generata aggiunta a questo volume. Le prove disponibili in Parte I e nell'appendice rimangono complete.